

TIANSHUI, China

WMO: 570060

Lat: 34.577N

Lon: 105.752E

Elev: 3750

StdP: 12.81

Time Zone: 8.00 (E08)

Period: 90-03

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	17.5	20.2	0.6	6.5	29.9	4.8	8.1	29.7	10.0	36.2	8.3	36.1	0.4	250	0.387

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	17.7	90.1	67.5	87.4	66.9	85.2	66.5	71.7	82.9	70.5	81.4	69.3	80.0	4.6	90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
68.6	120.8	76.9	67.2	115.1	76.0	65.9	109.8	75.2	38.2	83.0	37.1	81.4	36.0	80.0	77.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4)	9.8	8.0	7.1	11.8	95.0	3.9	3.2	9.0	97.3	6.7	99.2	4.5	101.0	1.6	103.3	
(5)				10.3	74.1	3.8	1.4	7.6	75.1	5.4	76.0	3.2	76.8	0.5	77.9	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	54.0	30.9	36.9	45.6	56.2	64.2	70.9	75.6	73.2	64.6	53.5	42.7	32.9	(6)
(7)		DBStd	16.27	4.37	5.36	6.42	7.03	5.82	4.34	4.37	4.74	5.91	5.65	6.03	4.74	(7)
(8)		HDD50	1940	592	368	167	21	1	0	0	0	32	228	531		(8)
(9)		HDD65	4912	1057	788	602	274	83	6	0	3	78	357	668	996	(9)
(10)		CDD50	3408	0	0	30	209	441	626	794	718	439	140	11	0	(10)
(11)		CDD65	905	0	0	0	11	58	183	329	256	67	1	0	0	(11)
(12)		CDH74	7132	0	0	6	125	541	1418	2771	1867	401	3	0	0	(12)
(13)		CDH80	2262	0	0	0	16	114	408	1036	596	92	0	0	0	(13)
(14)	Wind	WSAvg	2.2	1.6	2.2	2.8	2.8	2.6	2.4	2.5	2.5	2.0	1.5	1.5	1.5	(14)
(15)	Precipitation	PrecAvg	21.2	0.2	0.2	0.7	1.7	2.2	2.6	3.7	3.5	3.6	1.9	0.6	0.2	(15)
(16)		PrecMax	30.4	0.4	0.7	1.7	3.2	4.8	6.9	7.7	8.6	7.1	4.5	1.6	0.7	(16)
(17)		PrecMin	12.8	0.0	0.0	0.0	0.4	0.4	0.6	1.0	0.6	1.3	0.5	0.1	0.0	(17)
(18)		PrecStd	4.8	0.1	0.2	0.4	0.8	1.1	1.4	1.6	2.1	1.8	1.0	0.4	0.2	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	49.6	59.6	73.4	82.0	86.8	91.3	94.9	92.4	88.1	73.6	64.2	49.2	(19)
(20)			MCWB	37.6	43.7	52.6	60.3	61.1	65.7	69.3	68.1	64.0	56.7	50.9	38.7	(20)
(21)		2%	DB	45.3	53.7	66.3	77.4	83.1	87.2	90.8	88.8	82.7	69.1	60.3	46.4	(21)
(22)			MCWB	34.7	41.4	48.9	57.5	59.7	64.6	68.4	68.0	65.7	56.1	49.1	37.0	(22)
(23)		5%	DB	42.0	50.0	61.5	73.5	79.9	85.0	88.1	86.0	78.8	66.1	56.9	43.9	(23)
(24)			MCWB	33.0	38.6	47.2	55.1	59.0	64.4	68.2	68.0	64.5	55.1	47.2	35.2	(24)
(25)		10%	DB	39.0	46.6	57.2	69.4	76.6	82.2	85.6	83.2	75.2	62.8	53.2	41.2	(25)
(26)			MCWB	31.4	36.9	44.9	53.1	58.3	63.9	68.0	67.4	63.1	54.0	45.7	33.4	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	38.4	44.8	56.0	62.5	65.2	69.4	73.7	73.1	71.0	60.2	53.1	40.9	(27)
(28)			MCD B	47.8	56.3	69.4	77.3	78.6	82.1	87.6	84.2	79.7	67.5	61.3	47.3	(28)
(29)		2%	WB	35.5	42.4	50.4	59.7	63.2	67.8	72.1	71.5	68.2	58.3	50.5	37.9	(29)
(30)			MCD B	43.8	52.3	63.8	73.4	76.6	80.2	84.1	82.5	78.5	66.0	57.7	44.2	(30)
(31)		5%	WB	33.5	39.9	48.0	57.0	61.3	66.8	70.9	70.3	65.7	56.4	48.7	36.0	(31)
(32)			MCD B	41.1	48.6	59.3	71.3	74.7	79.2	82.2	80.9	75.6	63.4	55.3	42.4	(32)
(33)		10%	WB	31.7	37.7	46.1	54.7	59.9	65.8	69.8	69.1	63.9	54.8	46.3	34.2	(33)
(34)			MCD B	38.0	44.6	55.9	66.7	72.4	77.9	80.9	79.3	73.1	61.4	52.6	40.2	(34)
(35)	Mean Daily Temperature Range	MDBR	14.9	15.3	16.9	18.5	19.2	18.7	17.7	16.6	14.9	14.7	15.7	14.7	(35)	
(36)		5% DB	MCD BR	21.4	22.7	26.0	26.3	26.7	25.2	22.6	22.0	20.8	20.2	21.4	19.6	(36)
(37)			MCWBR	13.2	12.5	12.3	11.0	9.2	8.1	6.3	6.5	7.3	9.2	11.8	12.2	(37)
(38)		5% WB	MCD BR	19.7	21.1	23.3	22.6	21.8	19.7	17.4	17.6	17.8	17.6	19.7	16.8	(38)
(39)			MCWBR	12.5	12.0	11.6	10.5	8.9	7.8	6.0	6.0	7.3	8.7	11.4	10.9	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.472	0.541	0.588	0.588	0.565	0.547	0.549	0.588	0.551	0.532	0.475	0.432	(40)	
(41)		taud	1.790	1.677	1.626	1.664	1.784	1.882	1.935	1.810	1.853	1.821	1.858	1.904	(41)	
(42)		Ebn at Noon	215	215	220	229	237	241	239	227	227	216	213	220	(42)	
(43)		Edh at Noon	54	68	77	78	70	63	60	67	61	58	50	46	(43)	
(44)	All-Sky Solar Radiation	RadAvg	820	977	1299	1599	1692	1684	1606	1447	1054	915	817	771	(44)	
(45)		RadStd	79	143	118	149	142	118	207	171	102	131	122	96	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days				
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65	
(46)	Station Only	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(46)
(47)	Regional (6 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(47)

Nomenclature: See separate page